

FULLSTACK DEVELOPMENT

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DAY:3

TODAY TOPIC:

Compiler and interpreter

Features of java

Jvm (java virtual machine)

Jre (java run time environment)

Jdk (java development kit)

Integrated development kit

COMPILER AND INTERPRETER

A compiler converts the whole program into machine language at one time.
Example: C Compiler, Java Compiler.

An interpreter executes the program line by line and shows errors immediately.
Example: Python Interpreter.

Features of java

Java is an object oriented, secure, portable and platform independent language.
Example: A Java program can run on Windows, Linux and Mac

JVM (Java Virtual Machine)

JVM is the main engine that runs Java programs. It converts bytecode into machine code and allows Java to run on different platforms.

Main Functions of JVM

- Loads class files
- Verifies bytecode
- Executes code
- Manages memory
- Runs Garbage Collector

JRE (Java Runtime Environment)

JRE provides the environment required to run Java applications. It contains JVM and supporting libraries.

Components of JRE

- JVM
- Core libraries
- Supporting files

EX:

JRE = JVM + Libraries

JDK (Java Development Kit)

JDK is used to develop Java applications. It contains JRE and development tools.

Components of JDK

- JRE
- JVM
- Compiler
- Debugger
- Documentation tools

WORKFLOW

Jvm-runs java byte code

Jre-jvm+libraries(to run java apps)

Jdk-jre+tools(to develops java apps)

IDE (Integrated Development Environment)

IDE is a software application that helps programmers write, edit, compile, debug and run programs easily.

Main Features

- Code Editor
- Compiler
- Debugger
- Auto Completion
- Project Management

Example

A programmer writing Java code in a company needs JDK because they must:

- Write code
- Compile code
- Debug errors
- Run programs